

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



January 10, 2007.

MATERIALS LABORATORY STUDY

Report No. 06-092

A. ACCIDENT

Place : Boston, Massachusetts.
Date : July 10, 2006.
Vehicle : Connector Tunnel to Ted Williams Tunnel.
NTSB No. : HWY06MH024.
Investigator : Mark Bagnard.

B. COMPONENTS STUDIED

Void formation in a simulated adhesive anchor system hole.

C. DETAILS OF THE STUDY

On October 12, 2006, representatives from the National Transportation Safety Board's Materials Laboratory and Office of Highway Safety (NTSB) and the Federal Highway Administration's Turner Fairbank Highway Research Center (FHWA) conducted a series of tests in order to study void formation in an overhead adhesive anchor system. Participating in the tests were:

Derek Nash	NTSB Materials Lab
Jim Wildey	NTSB Materials Lab
Carl Schultheisz	NTSB Materials Lab
Dennis Collins	NTSB Office of Highway Safety
Benjamin Graybeal	FHWA
Justin Ocel	FHWA
Marshall Davis	FHWA

Background information

An adhesive anchor system was installed in the reinforced-concrete roof of the connector tunnel to Ted Williams Tunnel. Previous work (Materials Laboratory Report No. 06-073) revealed that the holes in the concrete roof of the tunnel had been drilled using a 3/4 inch diameter core drill. A core drill effectively drills out a ring, to the depth required, and the core is subsequently broken off to produce a hole. The previous work also

revealed that the core normally broke off approximately 1/4 inch above the bottom of the hole leaving a roughly flat bottom, approximately 0.6 inches diameter, which was encircled by a groove. The volume of the groove was calculated at 0.04 cubic inches (0.65 milliliters). The anchors used in the tunnel were 5/8-11 UNC¹ threaded rod produced from grade 316 stainless steel². The adhesive used was a Powers Fasteners two-part epoxy.

Creation of Simulated Core-Drilled Anchor Holes

To create a simulated anchor hole that would allow the anchor installation and void formation process to be directly observed, clear acrylic tubing with an inside diameter of 0.75 inch was cut into 5.5-inch lengths. A 0.25-inch high, 0.6-inch diameter disk of clear plastic was glued to the center of a 2 inch by 3 inch acrylic plate, and the acrylic plate was then glued onto one end of the acrylic tube such that the disk was approximately centered in the 0.75-inch diameter tube. Once assembled, the inner surface of the capped tubes simulated the approximate shape and size of a 3/4 inch diameter hole drilled with a coring bit, and the acrylic plate allowed the test hole to be clamped to an overhead plate simulating the orientation of the anchor holes in the tunnel roof. A photo of the upper portion of a test hole is illustrated in figure 1 with the acrylic tube, the clear plastic disk and the acrylic plate identified. A 5/8"-11 UNC nylon threaded rod, with a similarly chamfered end, was used as a substitute for the stainless steel anchor used in the tunnel. In order to observe what occurred during the test procedure a clear silicon-based caulk was used as a substitute for the opaque two-part epoxy used in the connector tunnel.

For comparison purposes, tests were also completed using Powers Fasteners "Quick Set" epoxy instead of the clear caulk. The tip of the injection tube supplied with the epoxy cartridge was flat (perpendicular to the axis of the nozzle). Additional mixing nozzles, available in packs of two, were also acquired and found to be of a one-piece construction (although the box illustrated a two piece construction) and the tip of the nozzle was cut at approximately 45 degrees. The length of the plain portion of the replacement nozzles could be reduced, if required, by snapping it off at two circumferentially notched locations, which would produce a flat tip.

The installation guidelines printed on the cartridge detail indicate that

- the anchor hole should be filled approximately half to two-thirds full, starting from the bottom or back of the anchor hole,
- the nozzle should be slowly withdrawn as the hole fills to avoid creating air pockets,
- the anchor should be turned slightly as it is inserted, and
- the anchor should be fully seated at the bottom of the hole.

¹ Unified National Coarse thread of 5/8 inch major diameter and 11 threads per inch.

² A stainless steel containing 16-18% chromium and 10-14% nickel as the major alloying elements.

Illustrations on the cartridge guidelines show a horizontal installation with holes drilled using a two-flute twist drill.

Testing

Prior to the testing, one of the simulated anchor holes was filled with water, and a nylon anchor was inserted. The water was allowed to overflow, and the anchor was extracted. The level of water remaining was approximately 2.5 inches deep, just short of the half way mark, and was used as a guide for the amount of caulk to be injected. A standard caulk gun was utilized to inject the silicon caulk and a length of rigid plastic tubing, with dimensions similar to those of the epoxy injection tube, was fastened to the tip of the silicon caulk cartridge.

The testing protocol consisted of labeling a test hole and clamping it to the overhead plate. Test holes were labeled alphabetically, starting with the letter "A", with a subsequent letter being used for each change in a test condition. The silicon caulk or epoxy was injected upwards into the test hole, and a nylon anchor was inserted. Caulk or epoxy injection was performed in accordance with the parameters stated for the following test holes. Anchors were inserted in one continuous motion with the speed of insertion controlled by the viscosity of the caulk and estimated at approximately 10 inches per minute. After testing, any voids produced were plotted on a sheet of graph paper, and an approximation of the void area was made. Small voids at the upper ends of the holes, indicated with yellow arrows in the post-test photos in figures 2 to 9, were not included in the area calculations.

"A" tests.

The "A" tests were attempts to produce what could be termed as an "ideal" anchor joint with the caulk filling the upper half of the test hole and no air entrapped in the top of the test hole prior to anchor insertion.

In test hole "A1" the caulk was injected through a nozzle with a 45-degree tip. In the post insertion photo illustrated in the left image of figure 2, a yellow arrow indicates a void on one side at the upper end of the test hole. During the last inch of anchor insertion, a small amount of trapped air from around the center clear plastic disk was seen to flow around the anchor to produce the void noted.

In test hole "A2", illustrated in the right image of figure 2, the caulk was injected through a nozzle with a flat tip. The yellow arrow indicates a void at the upper end of the hole that was produced by the flow during the last inch of insertion, as in test hole "A1". Also during the last inch of insertion, a void indicated by the green arrow in figure 2 formed on the side of the hole. The void was centered approximately 1.75 inches from the top of the test hole and encompassed an area of approximately 0.28 square inch.

“B” tests.

The “B” tests repeated the conditions of the “A2” test, but the caulk was injected with the tip of the nozzle starting at approximately 0.5 inch from the top of the hole to produce a significant air gap, approximately 0.75 inch long, in the top of the hole.

For test hole “B1”, the air gap is illustrated in the left image of figure 3 and indicated by the yellow arrow. The total volume of the air gap was calculated to be 0.26 cubic inch (4.26 milliliters) and consisted of 0.04 cubic inch for the 0.25 inch deep groove around the central disk and 0.22 cubic inch for the 0.5-inch long portion of the cylinder.³ After anchor insertion, a small void remained in the top of the hole, and longitudinally oriented voids were produced along the side of the hole. The right image in figure 3 illustrates the hole after anchor insertion, with a yellow arrow indicating a void that remained in the top of the hole. The upper green arrow indicates a void at the side of the hole similar to that produced in hole “A2”, and the lower green arrow indicates a longer void at the side of the hole that was approximately 3 1/8 inches long and intersected the bottom of the hole. Some of the air in the air gap at the beginning of the test escaped from the bottom of the test hole as the nylon anchor was inserted, producing a distinctive sound as it did so. The voids at the upper end and side of the hole, as before, were produced during the last inch of anchor insertion. Along the side of the hole, the upper void encompassed an area of approximately 0.19 square inch and the lower void had an area of approximately 1.1 square inches.

Test hole “B2” was a repeat of test hole “B1”. The total volume of the air gap was also calculated to be 0.26 cubic inches (4.26 milliliters). Insertion of the anchor produced a void at the top and a long void along the side of the hole that was similar to the long void in test hole “B1”. The approximately 3.5 inch-long void intersected the bottom of the hole and encompassed an area of approximately 1.4 square inches. The voids, as before, were produced during the last inch of anchor insertion. Due to its similarity with test hole “B1”, test hole “B2” is not illustrated.

Test hole “B3” was a repeat of test hole “B1” with the addition of a slight twist clockwise (as viewed looking upward) after anchor insertion. The left image of figure 4 illustrates the test hole prior to anchor insertion with the air gap indicated by the yellow arrow. The total volume of the air gap was calculated to be 0.26 cubic inch (4.26 milliliters). After anchor insertion (but before twisting of the anchor), a void was formed at the top of the test hole, and a longitudinal void similar to the lower void produced in test hole “B1” (illustrated in the right image in figure 3) was observed at the side of the hole. The voids, as before, were produced during the last inch of anchor insertion. After the slight twist, the longitudinal void separated into four distinct voids, and the effective area expanded. The right image in figure 4 illustrates the hole after anchor insertion with a yellow arrow indicating a bubble remaining in the top of the hole. The four green arrows

³ The upper surface of the caulk was uneven, and the distance from the disk to the surface of the caulk was estimated by estimating the height of an evened out surface.

indicate the four distinct voids produced by the twisting action. The approximate area of the voids produced, from top to bottom were, respectively, 0.4, 0.7, 0.7 and 1.13 square inches.

“C” test.

The “C” test repeated the conditions of the “A2” test, but the caulk injection was interrupted during the controlled extraction of the caulk gun in order to induce an air gap with caulk above it and below it.

Test hole “C1”, prior to anchor insertion, is illustrated in the left image of figure 5 with a yellow arrow indicating air trapped around the center disk at the top of the hole and a blue arrow indicating the induced air gap. The volume of the induced air gap was calculated at 0.22 cubic inch (3.6 milliliters). During the anchor insertion, most of the induced air gap was squeezed out of the test hole leaving only a small void centered at 4.25 inches from the top. The air at the top of the hole was relocated to one side of the hole appearing similar to the upper end void in test hole “A1” (indicated by the yellow arrow in the left image of figure 2). At this point, the anchor was given a slight twist and the test hole was re-identified as “C1a”. The twisting action spread out the voids at the top of the hole and the side. Test hole “C1a” is illustrated in the right image of figure 5 with a yellow arrow indicating the top void and a green arrow indicating the void at the side of the hole. The approximate area of the void at the side of the hole was 0.08 square inch. The red arrow in the right image of figure 5 indicates a longitudinally oriented area where a cluster of small isolated voids was observed in the roots of several anchor threads. The area was approximately 4 inches long and 0.2 inch wide.

“D” test.

The “D1” test repeated the conditions of the “A2” test, but the nozzle was not extracted any amount until after injection of the caulk. As the nozzle was extracted, some caulk adhered to the nozzle, and some was deposited on the sides of the hole below the half way mark. Test hole “D1” is illustrated in the left image of figure 6, prior to anchor insertion, with a blue arrow indicating the caulk deposited on the sides of the hole. During insertion of the anchor, caulk flowed around the anchor leaving a void at the top of the hole, a small void on the side of the hole, and a longitudinal band of small isolated voids in the thread roots of the anchor. The right image of figure 6 illustrates test hole “D1” after anchor insertion, with a yellow arrow indicating the void at the top of the hole, a green arrow indicating the small void at the side of the hole, and a red arrow indicating the location of the thread root void cluster. The approximate area of the side void was 0.03 square inch, and the band containing the thread root void cluster was approximately 1.7 inches long and 0.1 inch wide.

“E” tests.

The “E” tests repeated the conditions of the “A2” test, but a known quantity of air was injected after anchor insertion in one test hole and before anchor insertion in another test hole. The air was injected using a long needle on a graduated syringe.

In test hole “E1”, 6.0 milliliters of air were injected after anchor insertion. The injection needle was extracted during the air injection process. The left image of figure 7 illustrates the test hole, prior to anchor insertion, with a yellow arrow indicating the air trapped around the center disk at the top of the hole. The center image of figure 7 illustrates the test hole with the anchor inserted and the injector needle inserted. The yellow arrow indicates the void at the top of the hole, which appeared similar to the top void produced during tests “A1” and “A2” and indicated by the yellow arrows in figure 2. The blue arrow indicates the injector needle. The right image of figure 7 illustrates the test hole, after removal of the injector needle, with the green arrows indicating the longitudinal void that was produced at the side of the hole and the blue arrow indicating the thin extraction path of the needle after the 6.0 milliliters of air had been injected. The approximate area of the void produced was 0.87 square inch. The red arrow in the right image of figure 7 indicates a portion of an area where clusters of small voids were observed in the roots of the threads. The cluster of small voids in the thread roots was shaped similar to a riding boot approximately 2 inches long and encompassed an area of approximately 0.38 square inch.

In test hole “E2” 7.0 milliliters of air were injected into the top of the hole before anchor insertion. The left image of figure 8 illustrates the test hole, prior to air injection, with a yellow arrow indicating the air trapped around the center disk in the top of the hole. The center image of figure 8 illustrates the test hole, after air injection, with a yellow arrow indicating the air gap produced by the air injection. The blue arrow indicates the injection needle still in the hole after the air injection was completed. The right image of figure 8 illustrates the test hole, after anchor insertion, with a yellow arrow indicating the void at the top of the hole and green arrows indicating four distinct voids along the side of the hole that were produced during the last inch of anchor insertion. The approximate area of the four voids along the side of the hole, from top to bottom, were 0.14, 0.04, 0.23 and 0.23 square inch, a total of 0.64 square inch.

“F” tests.

The “F” tests repeated two of the previous conditions, but the Powers Fastener’s Power-Fast Plus, fast set epoxy adhesive was substituted for the silicon caulk. The epoxy cartridge, with the flat tipped nozzle supplied with it, was installed in the appropriate caulk gun. The mixing nozzle was purged, to ensure a correct mixing ratio of resin and hardener, before the epoxy was injected into the test hole.

Test hole “F1” was a repeat of test hole “B1” conditions with a significant air gap produced at the top of the hole prior to insertion of the threaded rod. The left image of figure 9 illustrates the test hole, prior to anchor insertion, with a yellow arrow indicating the

air gap produced at the top of the hole. The total volume of air above the epoxy was calculated at 0.32 cubic inch (5.2 milliliters). During the anchor insertion process, it was noted that the epoxy flowed around the anchor easier than the caulk, and a bubble of air escaped from the open end of the test hole during the last inch of insertion. The center image of figure 9 illustrates the test hole after the anchor had been inserted, with a yellow arrow indicating the void produced at the upper end of the hole. A red arrow indicates where the air bubble escaped, and blue arrows indicate two faint lines that appeared to connect to the edges of the bubble and continued up the test hole. Confirmation of voids in test hole "F1" is described later under "sectioning of the "F" test holes".

Test hole "F2" was a repeat of test hole "D1" conditions with the epoxy fully inserted without extracting the nozzle, the nozzle extracted, and then the anchor inserted. Prior to insertion of the anchor, the test hole appeared similar to test hole "D1", illustrated in the left image of figure 6, but the contents of the hole were gray. The right image of figure 9 illustrates the test hole after anchor insertion with a yellow arrow indicating the void produced at the upper end of the hole. A visual examination did not reveal any evidence of the presence of voids along the side of the hole. Confirmation of voids in test hole "F2" is described later under "sectioning of the "F" test holes".

Sectioning of the "F" test holes.

The "F" test holes were allowed to cure overnight before any sectioning was performed (the manufacturers literature specifies that, at 75°F, the epoxy requires a minimum of 2 hours to cure and 24 hours to achieve maximum strength). The outer surfaces of test holes "F1" and "F2" were scribed with a longitudinal line for orientation and were then cut at approximately 1 inch intervals starting from the top of the hole. The individual sections produced were additionally identified with the letters "a" to "f", starting from the top. The individual sections of test holes "F1" and "F2" are illustrated in figure 10 with the "up" direction indicated.

Examination of the sections produced from test hole "F1" revealed that two voids had been produced between the anchor and the inner surface of the test hole tube. The larger of the two voids intersected the open (lower) end of the hole, at the location indicated by the red arrow in figure 10, and terminated in section "F1c". The green arrows in the upper image of figure 11 indicate portions of the larger void. The smaller void was mostly contained within section "F1b" and is indicated by the blue arrow in the upper image of figure 11. To reveal the shape of the void within the pieces, a high intensity halogen light was placed against the threaded rod. The diffused light highlighted the shape of the void within the piece, allowing it to be plotted on the graph paper for area estimation. The larger void was approximately 3.5 inches long and extended over an area of 1.85 square inches. The smaller void was centered approximately 0.5 inch from the tip of the longer void and extended over an area of approximately 0.31 square inch.

A closer examination of the voids in test hole "F1" revealed that the epoxy had adhered to the inner surface of the test hole and to the surface of the nylon threaded rod. A closer view of the void in section "F1f" is illustrated in the lower image of figure 11 with

the anchor and the test hole tube identified. The red arrow indicates the epoxy on the inner surface of the hole and the blue arrow indicates the epoxy on the surface of the anchor.

Examination of the sections produced from test hole "F2" revealed that one void had been produced between the anchor and the inner surface of the test hole tube. The void was teardrop shaped and centered approximately 1 inch from the open end of the hole. The mating surfaces between sections "F2e" and "F2f" are illustrated in the left image of figure 12 with the void indicated by the green arrows. The void penetrated through the whole length of section "F2e" but did not extend into section "F2d". The void was approximately 1.2 inches long and extended over an area of approximately 0.28 square inch.

A closer examination of the void in test hole "F2" revealed that, as in test hole "F1", the epoxy had adhered to the anchor surface and to the inner surface of the test hole. The void in section "F2e" is illustrated in the right view of figure 12 with the anchor and the test hole tube identified. The red arrow indicates the epoxy on the inner surface of the hole and the blue arrow indicates the epoxy on the anchor surface.

Derek Nash
Mechanical Engineer

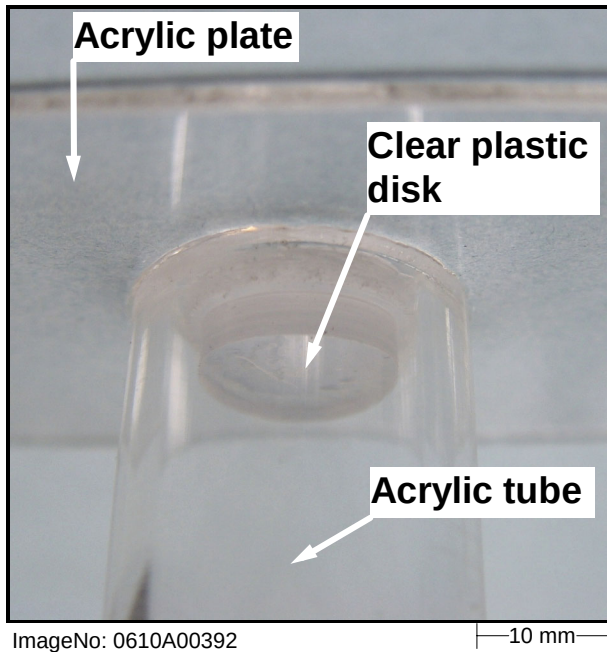


Figure 1. The typical construction at the upper end of the simulated test holes.

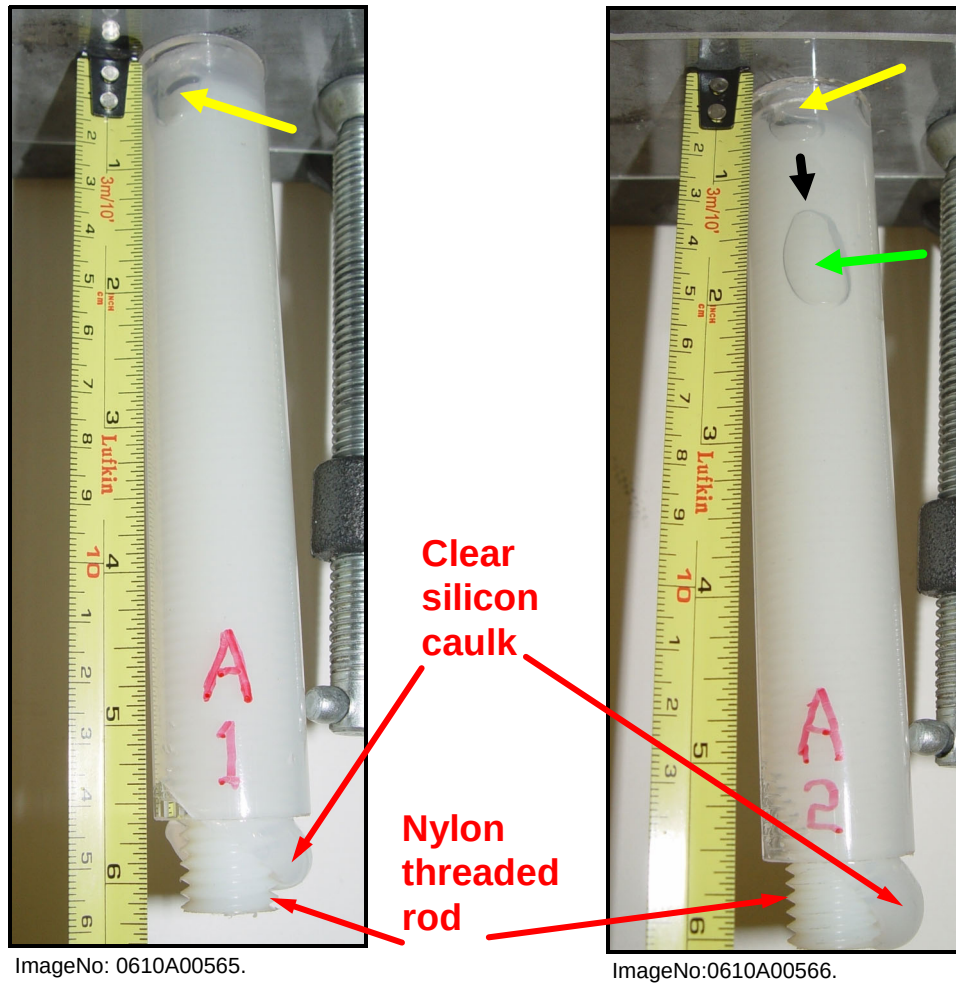
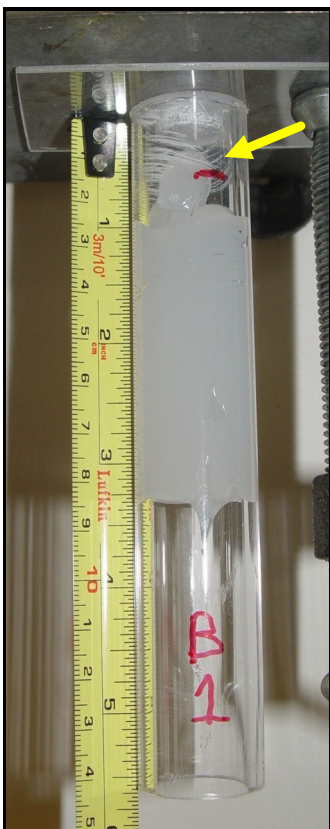


Figure 2. Test holes "A1" (left) and "A2" (right) after their respective test procedure.

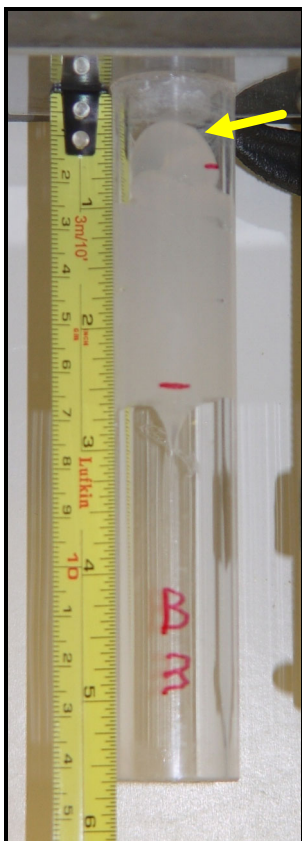


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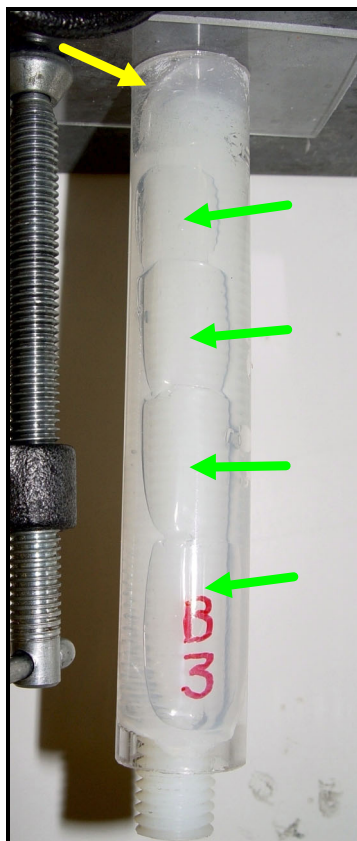


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Figure 3. Test hole "B1" before anchor insertion (left) and after anchor insertion (right).

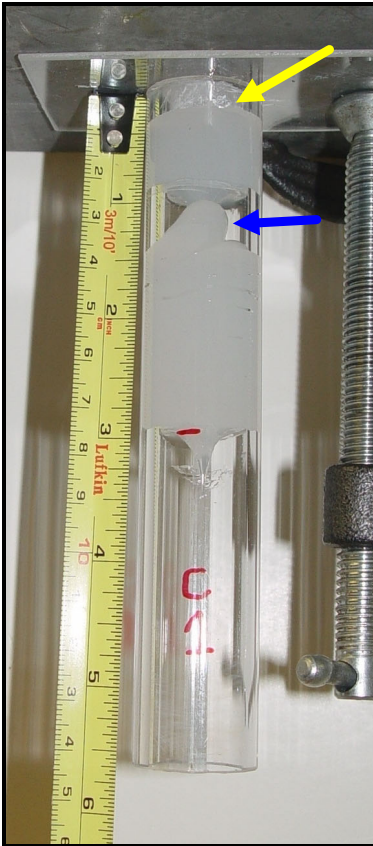


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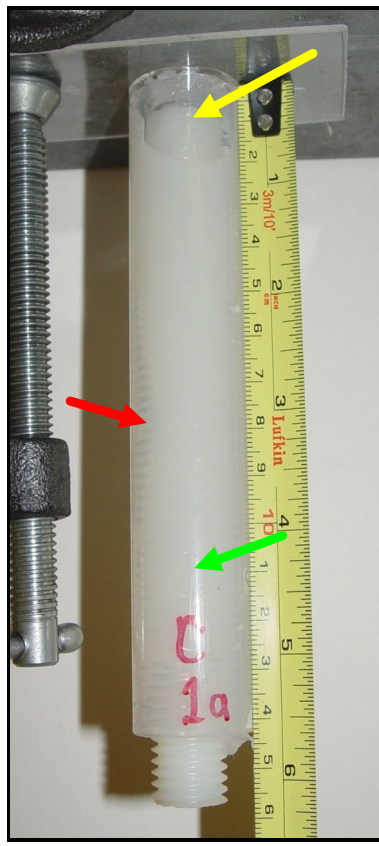


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Figure 4. Test hole "B3" before anchor insertion (left) and after anchor insertion (right).



ImageNo:0610A00575.

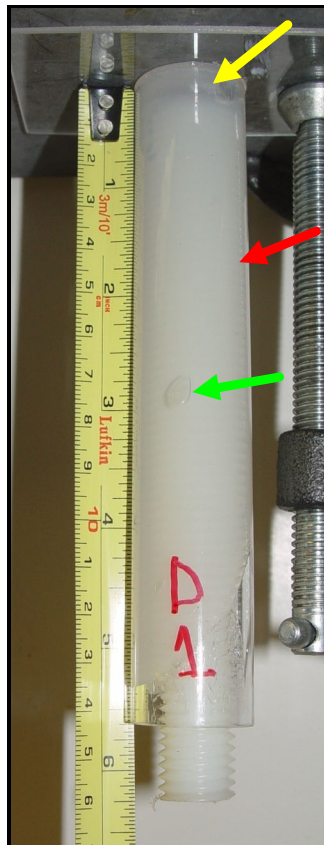


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Figure 5. Test hole "C1" before anchor insertion (left) and after anchor insertion with a slight twist (right).



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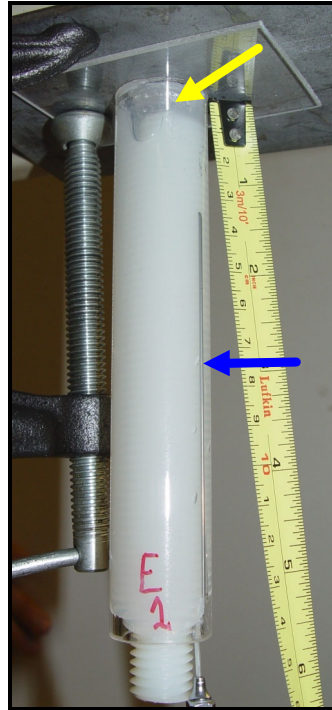


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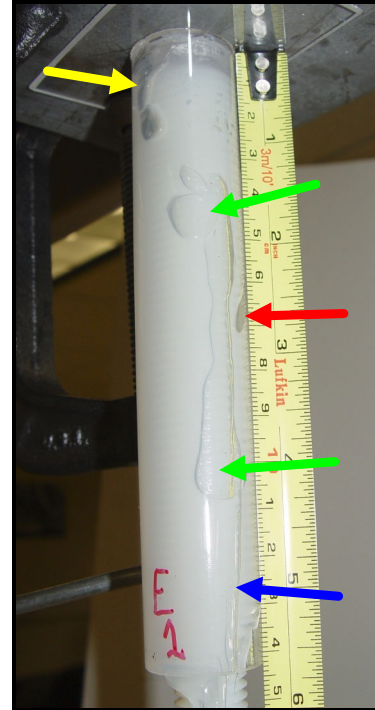
Figure 6. Test hole "D1" before anchor insertion (left) and after anchor insertion (right).



ImageNo:0610A00580.



ImageNo: 0610A00582.



ImageNo: 0610A00583

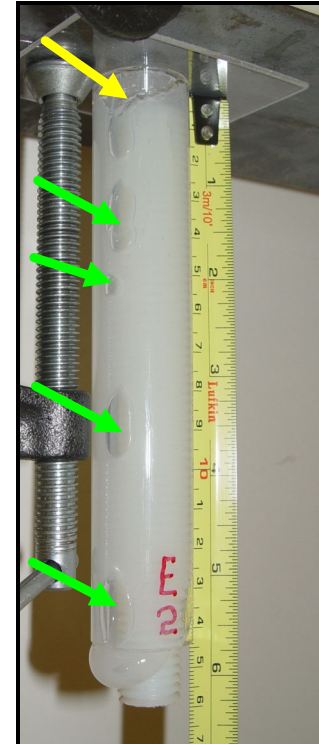
Figure 7. Test hole "E1" before anchor insertion (left), after anchor insertion followed by air injector needle insertion (center) and after air injection (right).



ImageNo: 0610A00584

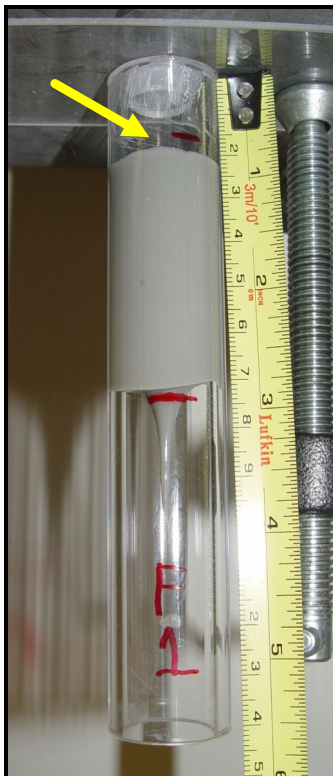


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ImageNo:0610A00586

Figure 8. Test hole "E2" before air injection (left), after air injection (center) and after anchor insertion (right).



ImageNo:0610A00587.

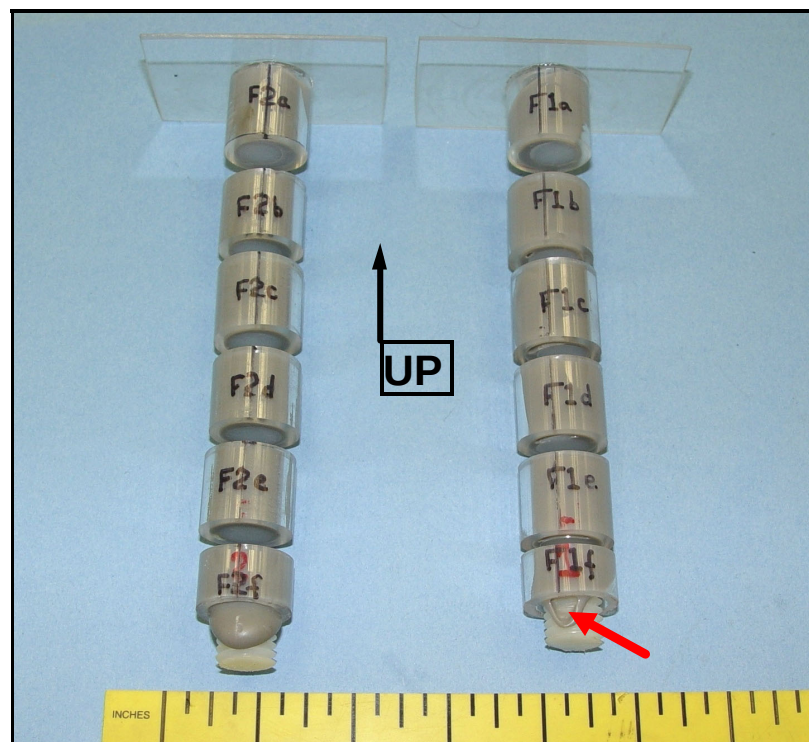


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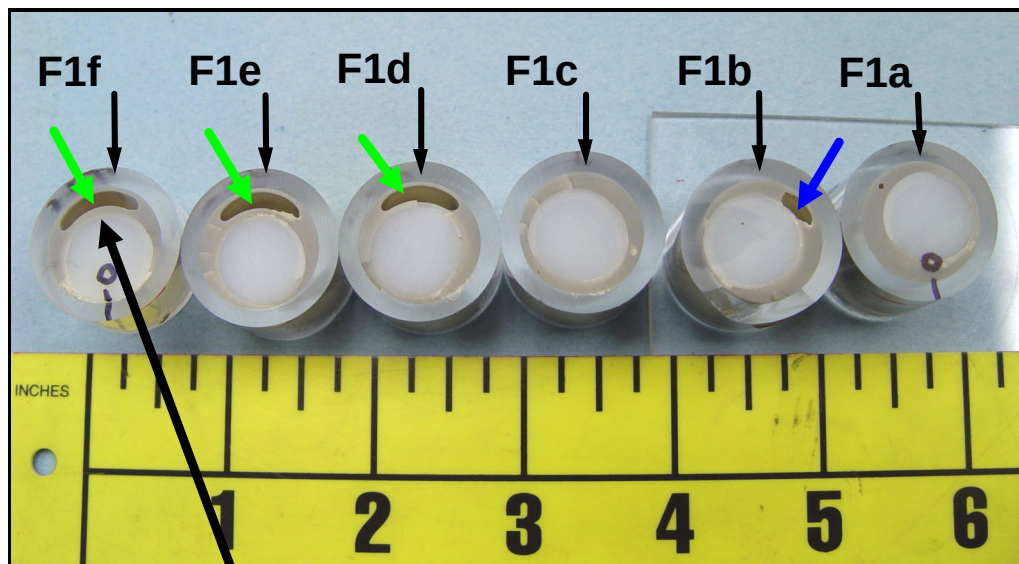
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Figure 9. Test hole "F1" before anchor insertion (left) and after anchor insertion (right). Test hole "F2" after anchor insertion (right).

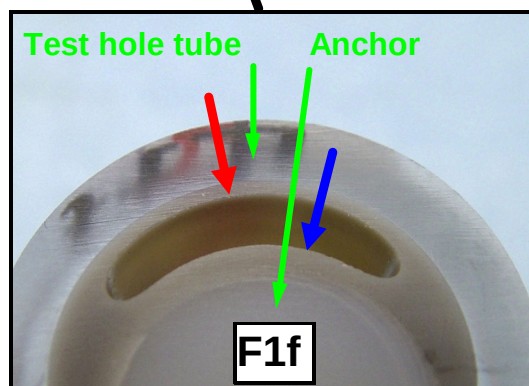


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Figure 10. Test holes "F1" (right) and "F2" (left) after being cut into sections "a" to "f".

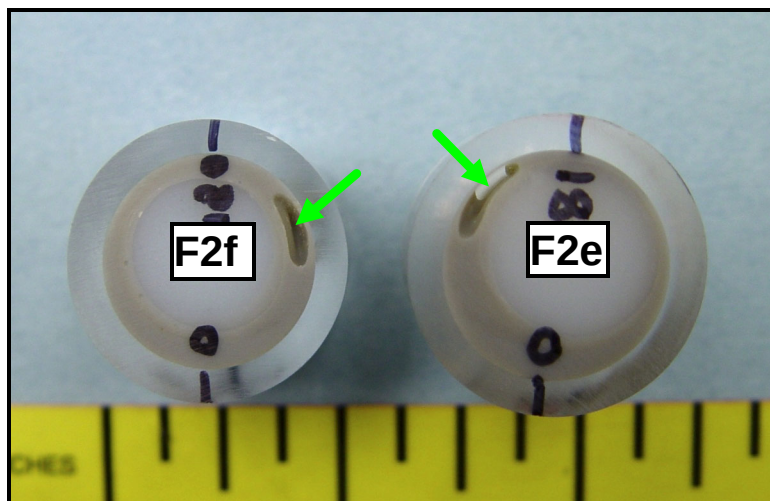


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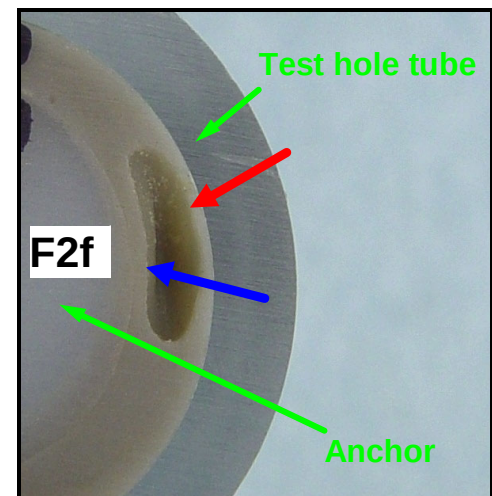


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Figure 11. Voids produced in test hole "F1" (top) and a closer view of the void on the upper surface of hole section "F1f" (left).



ImageNo: 0610A00639, Project No:2006090007



ImageNo: 0610A00638

Figure 12. Void produced in test hole "F2" (left) and a closer view of the void on the upper surface of hole section "F2f" (right).